

Aegis Quant Pricing Engine

Theoretical Background and Mathematical Derivations

Aegis Quant Library

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1 Introduction

This document provides the mathematical foundations underlying every model, analytical formula, and Monte Carlo simulation implemented in the Aegis quant pricing engine. Each section states the stochastic process, the probability measure, the key assumptions, and the derivation of the pricing formula, with explicit references to the C# source files where the corresponding implementation lives.

The library prices:

- Vanilla and exotic FX options (Garman-Kohlhagen, SABR, barriers, Asians, digitals, baskets)
- Vanilla and exotic equity options (Black-Scholes-Merton, barriers, Asians, digitals)
- SABR-based barrier options under the T-forward measure

2 Probability Measures and the Change-of-Numeraire Theorem

2.1 Numeraires and equivalent martingale measures

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$ be a filtered probability space satisfying the usual conditions. A *numeraire* is any strictly positive $\mathcal{F}_t$ -adapted price process $N(t)$.

Definition 2.1 (Equivalent martingale measure (EMM)). A probability measure $\mathbb{Q} \sim P$ is an EMM associated with numeraire N if every asset price $V(t)$ (after dividing by N) is a local $\mathbb{Q}$ -martingale:

$$\frac{V(t)}{N(t)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{V(T)}{N(T)} \mid \mathcal{F}_t \right].$$

Absence of arbitrage is equivalent (under integrability conditions) to the existence of at least one EMM.

2.2 The risk-neutral measure $\mathbb{Q}$

The most common choice of numeraire is the *domestic money-market account* $B(t) = \exp(\int_0^t r(s) ds)$ with r the short rate. The associated EMM is the *risk-neutral measure* $\mathbb{Q}$. Under a flat rate r :

$$V(0) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[V(T)].$$

For FX, the domestic money-market account B^d denominates prices in the domestic currency (e.g. USD); for equity it is the risk-free bank account. Code reference: [GbmOptionMCPricer.cs](#) (`DiscountFactor` computed as $e^{-r_d T}$).

2.3 The T-forward measure $\mathbb{Q}^T$

The *T-forward measure* (or *T-annuity measure*) uses the zero-coupon domestic bond $P(t, T)$ as numeraire. By the change-of-numeraire theorem, if $\mathbb{Q}$ is the risk-neutral measure:

$$\left. \frac{d\mathbb{Q}^T}{d\mathbb{Q}} \right|_{\mathcal{F}_T} = \frac{P(0, T)^{-1}}{P(T, T)^{-1}} \cdot \frac{B(0)}{B(T)} = \frac{P(0, T)^{-1}}{1} \cdot e^{-rT} = 1 \quad (\text{flat rates, } P(0, T) = e^{-rT}).$$

Under flat rates the two measures coincide *for terminal payoffs*, so

$$V(0) = P(0, T) \mathbb{E}^T[V(T)].$$

The key property is that any traded forward $F(t, T) = S(t)/P(t, T)$ is a $\mathbb{Q}^T$ -martingale. This makes the $\mathbb{Q}^T$ -measure the natural setting for pricing futures and forward-settled payoffs: no

drift term appears in the forward's SDE. Code reference: `FXSABRMCPricer.cs` (SABR simulated under $\mathbb{Q}^T$; `DiscountFactor` = $e^{-r_d T}$, `InitialForward` = $S_0 e^{(r_d - r_f)T}$).

3 Geometric Brownian Motion and the Itô Formula

3.1 The GBM SDE and its solution

The spot process S satisfies under $\mathbb{Q}$:

$$\frac{dS}{S} = \mu dt + \sigma dW^\mathbb{Q},$$

where $W^\mathbb{Q}$ is a $\mathbb{Q}$ -Brownian motion. Applying Itô's lemma to $X = \ln S$:

$$dX = d \ln S = \frac{1}{S} dS - \frac{1}{2S^2} (dS)^2 = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW^\mathbb{Q}.$$

Integrating:

$$S(T) = S(0) \exp \left[\left(\mu - \frac{1}{2}\sigma^2 \right) T + \sigma W^\mathbb{Q}(T) \right].$$

The $-\frac{1}{2}\sigma^2$ correction is the *Itô correction* (convexity adjustment). Under $\mathbb{Q}$ one sets $\mu = r_d - r_f$ (FX) or $\mu = r - q$ (equity).

3.2 Log-Euler discretization

Applying the exact formula over a time step $[t_i, t_{i+1}]$ of length Δt :

$$S(t_{i+1}) = S(t_i) \exp \left[\left(f_d(t_i, t_{i+1}) - f_c(t_i, t_{i+1}) - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_i \right], \quad Z_i \sim \mathcal{N}(0, 1),$$

where f_d, f_c are the instantaneous forward rates from the respective curves. This is *exact* for GBM (no discretization error), which is why the Aegis MC pricers use log-Euler rather than plain Euler.

Code reference: `GbmOptionMCPricer.cs`, method `SimulateSpotPath`.

4 The FX Forward Measure and the Driftless Forward

4.1 FX covered-interest parity

Let $S(t)$ be the spot FX rate (domestic per unit foreign, e.g. USD/EUR). The foreign money-market account $B^f(t) = e^{r_f t}$ is a domestic asset with price $S(t)B^f(t)$ in domestic currency. Absence of arbitrage requires:

$$F(0, T) = S(0) \frac{B^d(T)}{B^f(T)} = S(0) e^{(r_d - r_f)T}.$$

This is *covered-interest parity*. Code: `FXSABRMCPricer.cs` (`InitialForward`), and `GbmOptionMCPricer.cs` (implicit via drift $r_d - r_f$).

4.2 Derivation: the forward is driftless under $\mathbb{Q}^T$

Under the domestic risk-neutral measure $\mathbb{Q}^d$:

$$\frac{dS}{S} = (r_d - r_f) dt + \sigma dW^d.$$

The T-forward is $F(t, T) = S(t)/P^d(t, T)$. Under flat rates $P^d(t, T) = e^{-r_d(T-t)}$, so $F(t, T) = S(t)e^{r_d(T-t)} \cdot e^{-r_d(T-t)+r_f(T-t)} \dots$. More precisely, applying Itô's lemma to $F = S e^{(r_d-r_f)(T-t)}$ under $\mathbb{Q}^d$:

$$\begin{aligned} dF &= e^{(r_d-r_f)(T-t)} dS - (r_d - r_f) S e^{(r_d-r_f)(T-t)} dt \\ &= e^{(r_d-r_f)(T-t)} [(r_d - r_f) S dt + \sigma S dW^d] - (r_d - r_f) S e^{(r_d-r_f)(T-t)} dt \\ &= \sigma F dW^d. \end{aligned}$$

The drift cancels exactly. This confirms $F(t, T)$ is a $\mathbb{Q}^d$ -local martingale (and hence a $\mathbb{Q}^T$ -martingale by the equivalence of the two measures under flat rates). Under $\mathbb{Q}^T$:

$$\frac{dF}{F} = \sigma dW^T, \quad (\text{no drift term}).$$

This is why `FXSABRMCPricer.cs` sets the drift to zero for the forward process: “Under the T-forward measure, the forward F is a martingale — no drift term.”

4.3 Pricing FX options in each measure

The same European call price can be written in two equivalent ways:

$$\begin{aligned} C &= e^{-r_d T} \mathbb{E}^d[\max(S(T) - K, 0)] && (\text{risk-neutral on } S) \\ &= e^{-r_d T} \mathbb{E}^T[\max(F(T) - K, 0)] && (\text{T-forward on } F, \text{ same terminal value}) \end{aligned}$$

Both representations are used in Aegis:

- `GarmanKohlhagen.cs` uses the risk-neutral representation (on S).
- `FXSABRMCPricer.cs` and `FXSABRVanillaOptionMCPricer.cs` use the T-forward representation (on F).
- `BlackForward.cs` prices the T-forward payoff without discounting.

5 Garman-Kohlhagen Formula

5.1 Model

Under the domestic risk-neutral measure $\mathbb{Q}^d$:

$$\frac{dS}{S} = (r_d - r_f) dt + \sigma dW^d.$$

5.2 Formula derivation

By the risk-neutral pricing formula:

$$C = e^{-r_d T} \mathbb{E}^d[\max(S(T) - K, 0)].$$

Write $S(T) = S_0 \exp[(r_d - r_f - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$ with $Z \sim \mathcal{N}(0, 1)$. Let d_1, d_2 be defined by the boundary $S(T) = K$ in Z -space:

$$d_1 = \frac{\ln(S_0/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

A standard Gaussian integration (splitting the expectation into “asset term” and “cash term”) gives:

$$\boxed{C = S_0 e^{-r_f T} \Phi(d_1) - K e^{-r_d T} \Phi(d_2)},$$

$$P = K e^{-r_d T} \Phi(-d_2) - S_0 e^{-r_f T} \Phi(-d_1).$$

Code: `GarmanKohlhagen.cs, Price()`.

Edge case. At $\sigma = 0$ the path is deterministic, $S(T) = F = S_0 e^{(r_d - r_f)T}$, and the price collapses to the discounted intrinsic: $e^{-r_d T} \max(\phi(F - K), 0)$.

6 Black-Scholes-Merton Formula (Equity)

The equity model is structurally identical to Garman-Kohlhagen with the substitutions $r_f \rightarrow q$ (dividend yield) and $r_d \rightarrow r$:

$$\frac{dS}{S} = (r - q) dt + \sigma dW,$$

giving:

$$C = S_0 e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Code: `BlackScholes.cs, Price()`; `GbmOptionMCPricer.cs` (Monte Carlo path generation with $r_{\text{discount}} = r, r_{\text{carry}} = q$).

7 Black Forward Formula

Under the T-forward measure $\mathbb{Q}^T$ the forward satisfies $dF/F = \sigma dW^T$ (driftless), with $F(T) = F_0 \exp(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T} Z)$. The undiscounted European payoff:

$$C_{\text{fwd}} = \mathbb{E}^T[\max(F(T) - K, 0)] = F_0 \Phi(d_1^F) - K \Phi(d_2^F),$$

$$d_1^F = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2^F = d_1^F - \sigma\sqrt{T}.$$

The full price is $P(0, T) \cdot C_{\text{fwd}}$. Note d_1^F differs from GK's d_1 only by the forward substitution $F_0 \equiv S_0 e^{(r_d - r_f)T}$ and the absence of the rate terms inside the logarithm.

Code: `BlackForward.cs, Call(), Put()`. Used as the lognormal-SABR ($\beta = 1, \nu > 0$) benchmark in `FXSABRVanillaOptionMCPricerTests.cs`.

8 Bachelier (Normal) Model

The normal model replaces the lognormal GBM with arithmetic Brownian motion:

$$dF = \sigma_N dW^T,$$

so $F(T) = F_0 + \sigma_N \sqrt{T} Z$. The European option price is:

$$\boxed{C = P(0, T) [(F_0 - K) \Phi(d) + \sigma_N \sqrt{T} \phi(d)]},$$

$$P_{\text{bch}} = P(0, T) [(K - F_0) \Phi(-d) + \sigma_N \sqrt{T} \phi(d)],$$

where $d = (F_0 - K)/(\sigma_N \sqrt{T})$ and ϕ is the standard normal PDF.

This can be derived by direct Gaussian integration:

$$C = P(0, T) \int_{(K-F_0)/(\sigma_N \sqrt{T})}^{\infty} (F_0 + \sigma_N \sqrt{T} z - K) \phi(z) dz.$$

Splitting: $(F_0 - K)\Phi(-d_{\text{lower}}) + \sigma_N \sqrt{T} \phi(d_{\text{lower}})$ where $d_{\text{lower}} = -d$, and using $\phi(d) = \phi(-d)$ gives the stated formula.

Usage in Aegis. σ_N is the *normal volatility* (in forward units per $\sqrt{\text{year}}$). Code: [Bachelier.cs](#), `Price()`. Used as the analytical benchmark for normal-SABR ($\beta = 0$) in [FXSABRVanillaOptionMCPricerTests](#).

9 Digital Options

9.1 Cash-or-nothing

Pays a fixed amount Q if $S(T) > K$ (call) or $S(T) < K$ (put):

$$\text{CoN_Call} = e^{-rT} Q \Phi(d_2), \quad \text{CoN_Put} = e^{-rT} Q \Phi(-d_2).$$

This follows immediately from $\mathbb{E}^Q[\mathbf{1}_{S(T) > K}] = \Phi(d_2)$, which is the “cash term” of Black-Scholes.

9.2 Asset-or-nothing

Pays $S(T)$ if $S(T) > K$:

$$\text{AoN_Call} = S_0 e^{-qT} \Phi(d_1).$$

This is the “asset term” of Black-Scholes; equivalently, it equals the derivative of the call with respect to K (up to sign), giving the connection $\partial C / \partial K = -e^{-rT} \Phi(d_2)$ (call spread / cash digital).

Code: [DigitalBlackScholes.cs](#), `CashOrNothing()`, `AssetOrNothing()`.

10 Geometric Asian Options

10.1 Model

Monitoring at n equally-spaced dates $t_i = iT/n$, $i = 1, \dots, n$. Under $\mathbb{Q}$, $\ln S(t_i)$ is Gaussian:

$$\ln S(t_i) = \ln S(0) + \left(r - q - \frac{1}{2}\sigma^2\right)t_i + \sigma W(t_i).$$

10.2 Moments of the geometric average

The discrete geometric average is:

$$G = \exp\left(\frac{1}{n} \sum_{i=1}^n \ln S(t_i)\right).$$

Since $\ln S(t_i)$ are jointly Gaussian, $\ln G$ is also Gaussian.

Mean:

$$\mathbb{E}[\ln G] = \ln S(0) + \left(r - q - \frac{1}{2}\sigma^2\right) \frac{T}{n} \cdot \frac{n(n+1)/2}{n} = \ln S(0) + \left(r - q - \frac{1}{2}\sigma^2\right) T \frac{n+1}{2n}.$$

Variance:

$$\begin{aligned}\text{Var}[\ln G] &= \frac{\sigma^2}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(W(t_i), W(t_j)) \\ &= \frac{\sigma^2}{n^2} \sum_{i,j} \min(t_i, t_j) = \frac{\sigma^2}{n^2} \cdot \frac{T}{n} \cdot \frac{n(n+1)(2n+1)}{6} \\ &= \sigma^2 T \frac{(n+1)(2n+1)}{6n^2}.\end{aligned}$$

(using $\sum_{i=1}^n \sum_{j=1}^n \min(i, j) = n(n+1)(2n+1)/6$).

10.3 Reduction to Black formula

Since G is lognormal, the Asian price is exactly the Black formula on a “synthetic” forward F_G with effective vol σ_{eff} :

$$F_G = S(0) \exp\left(\mathbb{E}[\ln G] - \ln S(0) + \frac{1}{2} \text{Var}[\ln G]\right), \quad \sigma_{\text{eff}} = \sqrt{\text{Var}[\ln G]/T}.$$

$$\text{Asian_Call} = e^{-rT} \text{BlackFwd}(F_G, K, T, \sigma_{\text{eff}}).$$

Code: [GeometricAsian.cs](#), `Price()`. The arithmetic-average Asian (no closed form) is computed by Monte Carlo in [EQAsianOptionMCPricer.cs](#) and [FXAsianOptionMCPricer.cs](#).

11 Zero Curves and Forward Rates

11.1 Discount factor and zero rate

A zero-coupon bond with maturity T is priced:

$$P(0, T) = e^{-r(T) \cdot T},$$

where $r(T)$ is the continuously-compounded zero rate. The function `ZeroCurve.InterpolateRate` performs linear interpolation in the zero-rate space over the `Pillars` term structure.

Code: [ZeroCurve.cs](#), `InterpolateRate()`, `DiscountFactor()`.

11.2 Instantaneous forward rate

The instantaneous (continuously-compounded) forward rate between t_0 and t_1 :

$$f(t_0, t_1) = \frac{r(t_1) \cdot t_1 - r(t_0) \cdot t_0}{t_1 - t_0}.$$

This ensures: $e^{-r(t_0)t_0} \cdot e^{-f(t_0, t_1)(t_1 - t_0)} = e^{-r(t_1)t_1}$, i.e. the forward discount is consistent with the zero curve.

Code: [ZeroCurve.cs](#), `ForwardRate()`. Used in [GbmOptionMCPricer.cs](#) to compute the per-step drift from the full curve:

$$\mu_i \cdot \Delta t = [f_d(t_i, t_{i+1}) - f_c(t_i, t_{i+1}) - \frac{1}{2}\sigma^2] \Delta t.$$

12 Discrete Dividends: the Escrowed Jump Model

12.1 Model

A cash dividend D_i is paid on ex-date $\tau_i \in (0, T]$. The equity spot drops by the dividend amount at τ_i (the *jump* or *escrowed* model):

$$S(\tau_i^+) = S(\tau_i^-) - D_i.$$

Combined with the continuous GBM drift:

$$S(t_{k+1}) = S(t_k) \exp[\mu_k \Delta t + \sigma \sqrt{\Delta t} Z_k] - D_k,$$

where $D_k = \sum_{\tau_i \in (t_k, t_{k+1}]} D_i$ is the sum of all dividends ex-dating during step k .

12.2 Validation

The model requires $\sum_{i: \tau_i \leq T} D_i < S(0)$ (otherwise the forward becomes negative in the deterministic limit). Code: `GbmOptionMCPricer.cs`, `PrepareDividends()`.

13 The SABR Stochastic Volatility Model

13.1 Model dynamics (T-forward measure)

Hagan, Kumar, Lesniewski, Woodward (2002). Under the T-forward measure $\mathbb{Q}^T$:

$$dF = \sigma C(F) dW_1^T, \tag{1}$$

$$d\sigma = \nu \sigma dW_2^T, \tag{2}$$

$$dW_1^T dW_2^T = \rho dt. \tag{3}$$

The CEV function $C(F)$ controls the backbone:

- $\beta = 0$: $C(F) = 1$ (normal/additive diffusion, forward can be negative)
- $\beta = 1$: $C(F) = F$ (lognormal/multiplicative diffusion)
- $0 < \beta < 1$: $C(F) = F^\beta$ (CEV, absorbing boundary at $F = 0$)

Initial conditions: $F(0) = F_0 = S_0 \exp[(r_d - r_f)T]$, $\sigma(0) = \alpha$. The option price is:

$$V = P(0, T) \mathbb{E}^T[\max(\phi(F(T) - K), 0)], \quad \phi = \pm 1.$$

Code: `FXSABRMCPricer.cs` (abstract base); all parameters in `SabrParameters` (protobuf).

13.2 Why the forward is a martingale under $\mathbb{Q}^T$

Since $C(F) \geq 0$ and the SDE (1) has no drift term, F is a local $\mathbb{Q}^T$ -martingale by construction (the T-forward measure was chosen precisely for this purpose; see Section 4).

13.3 Euler–Maruyama and log-Euler discretization

13.3.1 Volatility process (log-Euler, exact)

Since σ follows lognormal dynamics $d\sigma = \nu \sigma dW_2^T$, the exact solution over $[t, t + \Delta t]$ is:

$$\sigma(t + \Delta t) = \sigma(t) \exp[\nu \sqrt{\Delta t} \tilde{W}_2 - \frac{1}{2} \nu^2 \Delta t],$$

where $\tilde{W}_2$ is a standard normal. Log-Euler for σ is exact because the SDE is linear-multiplicative in σ with no F -dependence.

Code: `FXSABRMCPricer.cs`, `SimulateSabrPath()`.

13.3.2 Brownian decomposition

Two independent factors $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ from the simulation cube are combined to reproduce the correlation ρ :

$$\tilde{W}_1 = Z_1, \quad \tilde{W}_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2.$$

This factorization keeps ρ as a pure model parameter; bumping ρ never requires regenerating the cube.

Code: [FXSABRMCPricer.cs](#), field `_rhoComplement =  $\sqrt{1 - \rho^2}$` .

13.3.3 Forward process

$\beta = 1$ (lognormal, log-Euler, exact for frozen σ):

$$F(t + \Delta t) = F(t) \exp\left[-\frac{1}{2}\sigma^2\Delta t + \sigma\sqrt{\Delta t} Z_1\right].$$

No discretization error for constant σ ; cross-variation error with σ is $O(\Delta t)$.

$\beta = 0$ (normal, Euler-Maruyama, exact for constant σ):

$$F(t + \Delta t) = F(t) + \sigma\sqrt{\Delta t} Z_1.$$

$0 < \beta < 1$ (CEV, Euler with absorbing boundary):

$$F(t + \Delta t) = \max(F(t) + \sigma \max(F(t), 0)^\beta \sqrt{\Delta t} Z_1, 0).$$

Code: [FXSABRMCPricer.cs](#), method `StepForward()`.

14 Hagan SABR Implied Volatility Approximations

14.1 Lognormal (Black) implied volatility (β general)

Hagan et al. (2002) derived an asymptotic expansion for the lognormal (Black) implied vol $\sigma_B(F, K)$ accurate to $O(T)$. Define:

$$x = \ln(F/K), \quad \zeta = \frac{\nu}{\alpha} (FK)^{(1-\beta)/2} x, \quad \chi(\zeta) = \ln\left(\frac{\sqrt{1 - 2\rho\zeta + \zeta^2} + \zeta - \rho}{1 - \rho}\right).$$

The implied Black vol is:

$$\sigma_B(F, K) = \frac{\alpha}{(FK)^{(1-\beta)/2} D(x)} \cdot \frac{\zeta}{\chi(\zeta)} \cdot [1 + \mathcal{C} \cdot T],$$

where

$$D(x) = 1 + \frac{(1 - \beta)^2}{24} x^2 + \frac{(1 - \beta)^4}{1920} x^4,$$

$$\mathcal{C} = \frac{(1 - \beta)^2 \alpha^2}{24(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{(1-\beta)/2}} + \frac{(2 - 3\rho^2)\nu^2}{24}.$$

Limiting cases:

- $\nu \rightarrow 0$: $\zeta \rightarrow 0$, $\zeta/\chi \rightarrow 1$; reduces to CEV vol $\alpha/(FK)^{(1-\beta)/2}$.
- $F \rightarrow K$ (ATM): $x \rightarrow 0$, $D \rightarrow 1$; no 0/0 if handled by L'Hôpital.
- $\beta = 1$, ATM: $\sigma_B \rightarrow \alpha(1 + \mathcal{C}T)$ with $\mathcal{C} = (\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24)$.

Code: [Hagan.cs](#), `BlackVol()`.

14.2 Normal implied volatility ($\beta = 0$, ATM)

The Hagan ATM normal-vol formula:

$$\sigma_N(F, F) = \alpha \left(1 + \frac{(2 - 3\rho^2)\nu^2}{24} T \right).$$

Combined with the Bachelier formula (Section 8), this benchmarks the normal-SABR ($\beta = 0$) MC prices.

Code: [Hagan.cs](#), `NormalVolAtm()`.

15 Barrier Options

15.1 Definitions

A single-barrier option pays a vanilla payoff $\max(\phi(S(T) - K), 0)$ only if the barrier H is (or is not) crossed by the process during $[0, T]$:

Type	Activated by	If not activated
Up-and-In	$S(t) \geq H$ for some t	rebate
Up-and-Out	$S(t) < H$ for all t	rebate
Down-and-In	$S(t) \leq H$ for some t	rebate
Down-and-Out	$S(t) > H$ for all t	rebate

The model-free identity for zero rebate:

$$V_{\text{KI}} + V_{\text{KO}} = V_{\text{vanilla}},$$

holds path-by-path, providing an exact numerical sanity check (tested in [FXBarrierOptionMCPricerTests.cs](#) and [FXSABRBarrierOptionMCPricerTests.cs](#)).

15.2 Discrete monitoring

The barrier is checked at each MC time step t_k . Code: [FXBarrierOptionMCPricer.cs](#), `CheckDiscreteBarrier()`, [FXSABRBarrierOptionMCPricer.cs](#), `CheckDiscreteBarrier()`.

15.3 Continuous monitoring: Brownian bridge correction

With discrete time steps the MC pricer may miss barrier crossings occurring between observation dates. The Brownian bridge correction (Baldi, Caramellino, Iovino 1999) estimates the conditional probability that a continuous GBM path crosses H in $[t_k, t_{k+1}]$, given the endpoints $a = S(t_k)$ and $b = S(t_{k+1})$ on the *same side* of H :

$$P(\text{cross} \mid a, b, H) = \exp\left(\frac{-2 \ln(H/a) \ln(H/b)}{\sigma^2 \Delta t}\right).$$

15.3.1 Derivation

Under the T-forward measure, the log-process $X(t) = \ln F(t)$ is a Brownian motion with diffusion σ (for $\beta = 1$). Conditioned on $X(t_k) = \ln a$ and $X(t_{k+1}) = \ln b$, the bridge $\{X(s) : t_k \leq s \leq t_{k+1}\}$ is a Brownian bridge with variance $\sigma^2(s - t_k)(t_{k+1} - s)/(t_{k+1} - t_k)$.

The maximum of a Brownian bridge B^{br} on $[0, 1]$ with $B^{\text{br}}(0) = u$, $B^{\text{br}}(1) = v$ satisfies:

$$P\left(\sup_{s \in [0,1]} B^{\text{br}}(s) \geq h\right) = \exp(-2h(h - u - v)) \mathbf{1}_{h > \max(u,v)}.$$

Translating to the log barrier $h = \ln H$, $u = \ln a$, $v = \ln b$ and rescaling by total variance $\sigma^2 \Delta t$:

$$P = \exp\left(-2 \cdot \frac{(\ln H - \ln a)(\ln H - \ln b)}{\sigma^2 \Delta t}\right) = \exp\left(\frac{-2 \ln(H/a) \ln(H/b)}{\sigma^2 \Delta t}\right).$$

When $\ln(H/a)$ and $\ln(H/b)$ have opposite signs (endpoints on opposite sides), $P = 1$ (certain crossing).

Code references:

- `FXBarrierOptionMCPricer.cs`: `BridgeCrossingProb()`, `CheckContinuousBarrier()` with $\sigma^2 \Delta t = \text{SigSqDt}$ (constant vol).
- `FXSABRBarrierOptionMCPricer.cs`: same formula, with $\sigma^2 \Delta t$ replaced by the stochastic *beginning-of-step* SABR vol:

$$\sigma^2 \Delta t \rightarrow v_{\text{beg}}(t)^2 \Delta t,$$

where $v_{\text{beg}}(0) = \alpha$ and $v_{\text{beg}}(t_k) = \text{volPath}[k-1]$. This is exact for $\beta = 1$ (locally lognormal) and an approximation for other β .

15.3.2 Reproducibility

Bridge draws use a deterministic seed per (path, step):

```
var pathSeed = CurrentPathIndex ^ unchecked((int)0x9E3779B9u);
var u = new Random(pathSeed ^ (t * 1000003)).NextDouble();
if (u < pCross) return true;
```

The salt `0x9E3779B9` (the fractional part of the golden ratio in 32-bit hex) distributes the seed space.

15.4 SABR barrier: barrier on the forward

FX barrier options are quoted on the spot S . The SABR MC pricer simulates the forward $F(t, T)$ (not S), so the barrier is expressed in forward space. Under the T-forward measure with flat rates:

$$F(t, T) = S(t) e^{(r_d - r_f)(T-t)}.$$

The barrier on F coincides with the barrier on S if and only if $r_d = r_f$ (then $F(t, T) = S(t)$ for all t). This is used in the degeneracy test `ZeroVolOfVol_EqualRates_SabrBarrier_MatchesGbmBarrier` in `FXSABRBarrierOptionMCPricerTests.cs`.

Code: `FXSABRBarrierOptionMCPricer.cs`, `EvaluatePayoff()`.

16 Monte Carlo Pricing

16.1 Estimator

Given M independent paths, the Monte Carlo price estimator is:

$$\hat{V} = \frac{1}{M} \sum_{m=1}^M \text{Payoff}_m, \quad \text{Var}[\hat{V}] = \frac{s^2}{M}, \quad s^2 = \frac{1}{M-1} \sum_m (\text{Payoff}_m - \hat{V})^2.$$

A 95% confidence interval is $\hat{V} \pm 1.96 s / \sqrt{M}$.

Code: `MCBasePricer.cs`, `Price()`; `PricingResult` carries `Price`, `StandardError`, `ConfidenceIntervalLower`

16.2 Parallel path evaluation

`MCBasePricer.Price()` uses `Parallel.For` to distribute paths across CPU cores with thread-local accumulators (sum, sum-of-squares), merged at the end. Thread safety is achieved via:

- ThreadLocal spot/forward buffers in `GbmOptionMCPricer.cs` and `FXSABRMCPricer.cs`.
- The `SimulationCube` is read-only after construction.

16.3 Simulation cube: pre-generated standard normals

The `SimulationCube` (shape $M \times \text{Steps} \times \text{Assets}$) pre-generates all required $\mathcal{N}(0,1)$ draws. For SABR, two assets are required (one for the forward, one for the independent vol factor). For single-asset GBM, one asset suffices.

Code: `SimulationCube.cs`.

16.4 Common random numbers (variance reduction)

Bumped pricers for Greeks reuse the same `SimulationCube` (`CreateBumped` returns a new pricer instance pointing to the same cube). The base price and the bumped price share path realizations, reducing the variance of the finite-difference estimator by positive path-level correlation.

Code: `FXSABRVanillaOptionMCPricer.cs` and `FXSABRBarrierOptionMCPricer.cs`, method `CreateBumped()`.

17 Greeks

17.1 Bump-and-reprice

All Greeks are computed by finite-difference bump-and-reprice:

$$\Delta = \frac{V(S + \varepsilon) - V(S - \varepsilon)}{2\varepsilon}, \quad \Gamma = \frac{V(S + \varepsilon) - 2V(S) + V(S - \varepsilon)}{\varepsilon^2}, \quad \nu = \frac{V(\sigma + \varepsilon) - V(\sigma - \varepsilon)}{2\varepsilon}.$$

For spot/rate bumps the same pricer instance is re-priced with bumped market data (`MarketDataBumps.cs`). For vol bumps, a new pricer instance is constructed with bumped σ or α , reusing the same cube.

17.2 SABR-specific Greeks

Vega ($\partial V / \partial \alpha$): bump $\alpha \rightarrow \alpha \pm \varepsilon$, construct new pricer, reprice on same market. α is the SABR initial vol level, playing the role of ATM vol.

Vol-of-vol sensitivity ($\partial V / \partial \nu$): bump ν , construct new pricer. Code: `FXSABRMCPricer.cs`, `ComputeVolOfVolSensitivity()`.

Correlation sensitivity ($\partial V / \partial \rho$): bump ρ (clamped to $(-1, 1)$), construct new pricer. Code: `FXSABRMCPricer.cs`, `ComputeCorrelSensitivity()`.

Theta: not computed (returned as NaN) because bumping time would require rebuilding the cube or re-parameterising the contract.

18 Basket Options

18.1 Model

An n -leg basket with weights w_i and a weighted-sum aggregation:

$$B(t) = \sum_{i=1}^n w_i S_i(t).$$

Each leg S_i follows correlated GBM under $\mathbb{Q}$:

$$\frac{dS_i}{S_i} = \mu_i dt + \sigma_i dW_i, \quad dW_i dW_j = \rho_{ij} dt.$$

The simulation uses a Cholesky decomposition of the correlation matrix to generate correlated standard normals from independent draws.

Code: `FXBasketVanillaMCPricer.cs`, `FXBasketBarrierOptionMCPricer.cs`.

18.2 Effective vol for basket barrier bridge

For continuous barrier monitoring on a basket, the effective basket volatility σ_{eff} at each step is reconstructed from the realized leg spots and the leg vols, used as the local diffusion coefficient in the Brownian bridge formula (Section 15).

19 Normal Distribution Utilities

The standard normal CDF is evaluated using the complementary error function:

$$\Phi(x) = \frac{1}{2} \operatorname{erfc}(-x/\sqrt{2}),$$

via `MathNet.Numerics.SpecialFunctions.Erfc`, giving full double-precision accuracy (no polynomial approximation error).

The PDF:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Code: `NormalDistribution.cs`, `Cdf()`, `Pdf()`.

20 Summary of Code-to-Formula Mapping

File	Model / Formula	Section
GarmanKohlhagen.cs	GK call/put	5
BlackScholes.cs	BSM call/put	6
BlackForward.cs	Black forward call/put	7
Bachelier.cs	Normal model call/put	8
DigitalBlackScholes.cs	Cash/asset-or-nothing	9
GeometricAsian.cs	Kemna-Vorst geometric Asian	10
Hagan.cs	SABR σ_B, σ_N	14
NormalDistribution.cs	Φ, ϕ via erfc	19
ZeroCurve.cs	Interpolation, $P(0, T), f(t_0, t_1)$	11
GbmOptionMCPricer.cs	GBM log-Euler, discrete dividends	3,12
FXSABRMCPricer.cs	SABR under $\mathbb{Q}^T$, Euler-Maruyama	13
FXBarrierOptionMCPricer.cs	Barrier, Brownian bridge (GBM)	15
FXSABRBarrierOptionMCPricer.cs	Barrier, Brownian bridge (SABR)	15
FXBasketVanillaMCPricer.cs	Correlated GBM basket	18
FXBasketBarrierOptionMCPricer.cs	Basket barrier + bridge	18,15
MCBasePricer.cs	MC estimator, confidence interval	16

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